Majid Daliri

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Research Profile

NYU Computer Science PhD student and **Apple Scholars in AI/ML Fellow (2025 - 2027)**. Research in efficient machine learning, vector quantization, randomized algorithms, and retrieval, with industry experience in AI agents and model optimization.

Education

New York University — New York, NY
Ph.D. in Computer Science — 2022 - Present
Advisor: Christopher Musco | Theoretical Computer Science Group
M.S. in Computer Science — 2022 - 2024

University of Tehran — Tehran, Iran
B.S. in Computer Engineering | GPA: 3.97/4.0 — 2017 - 2022

Industry Research Experience

Research Intern, Microsoft Research — Present
Research mentor: Lester Mackey

- Developing efficient **causal-attention algorithms** for YOCO architectures and diffusion models to reduce computational cost and accelerate model execution.

Research Intern, Google — Summer 2026
Research mentor: Jieming Mao

- Refined **agentic memory** for Google's Antigravity and Teamwork products.
- Created **datasets and benchmarks** for realistic financial tasks; evaluated agent accuracy, robustness, and task success.

Student Researcher, Google — Spring 2026
Research mentor: Rajesh Jayaram

- Developed **SliceTurbo**, a sliced variant of TurboQuant that uses **quantized filtering** to accelerate similarity-search pipelines.
- Built **benchmarks and ablation studies** for nearest-neighbor search, k-means clustering, and multi-vector search, measuring recall, clustering quality, latency, and memory tradeoffs.

Research Scientist Intern, Adobe — Summer 2025
Research mentor: Tung Mai

- Developed **quantization methods** for on-device models under tight memory and compute budgets.
- Built **quantization-aware training workflows** with ONNX and TorchScript integration for mobile and edge deployment.

Engineering Experience

Site Reliability Engineer, Cafebazaar — 2021 - 2022

Designed, implemented, and maintained managed Redis and PostgreSQL services on a Kubernetes-based cloud. Tools included Docker, Sentry, S3, and Prometheus.

Selected Honors

Apple Scholars in AI/ML Fellowship — **2025 - 2027**
One of 21 scholars selected for the 2025 cohort; full graduate fellowship.

Hong Kong PhD Fellowship Scheme, 2022; declined to attend NYU.
ACM ICPC Regional Contest, 2019; ranked 6th among more than 100 teams across Iran.

Selected Publications

- **On Worst-Case Guarantees for Graph-Based Nearest-Neighbor Search**
  Majid Daliri, Mohammadreza Daneshvaramoli, Purna Dutta, Edo Liberty, Cameron Musco, Christopher Musco, Torsten Suel
  NeurIPS 2026
  Paper contribution: Worst-case bounds for graph search and guarantees using Steiner points.
- **ATLAS: Learning to Optimally Memorize the Context at Test Time**
  Ali Behrouz, Zeman Li, Praneeth Kacham, **Majid Daliri**, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, Vahab Mirrokni
  ICML 2026
  Paper contribution: Long-term memory that learns from current and past context at test time.
- **TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate**
  Amir Zandieh, **Majid Daliri**, Majid Hadian, Vahab Mirrokni
  ICLR 2026
  Paper contribution: Near-optimal distortion guarantees for online vector quantization.
- **Coupling without Communication and Drafter-Invariant Speculative Decoding**
  Majid Daliri, Christopher Musco, Ananda Theertha Suresh
  ISIT 2025
  Paper contribution: Speculative decoding with fixed-seed outputs independent of the drafter.
- **QJL: 1-Bit Quantized JL Transform for KV Cache Quantization with Zero Overhead**
  Amir Zandieh, **Majid Daliri**, Insu Han
  AAAI 2025
  Paper contribution: Over 5x KV-cache memory reduction at 3-bit precision in evaluated LLMs.
- **Matrix Product Sketching via Coordinated Sampling**
  Majid Daliri, Juliana Freire, Danrong Li, Christopher Musco
  ICLR 2025
  Paper contribution: Smaller sketches for sparse matrix products through coordinated sampling.
- **KDEformer: Accelerating Transformers via Kernel Density Estimation**
  Amir Zandieh, Insu Han*, **Majid Daliri***, Amin Karbasi (* equal contribution)
  ICML 2023
  Paper contribution: Subquadratic attention approximation with spectral-norm error guarantees.

Full publication list and paper links: majid-daliri.github.io/articles/

Academic Research Experience

Research Intern, University of Salzburg Summer 2022
Research mentor: Ana Sokolova
Developed algorithms for **distribution bisimilarity**; investigated probabilistic system verification and quantum annealing for program verification.

Machine Learning Intern, Max Planck Institute 2021 - 2022
Research mentor: Amir Zandieh
Implemented fast attention methods for Transformers and designed a GPU-compatible locality-sensitive hashing method for **attention approximation**.

Research Intern, HKUST 2021 - 2022
Research mentor: Amir Goharshady
Developed approximation algorithms for **cache-conscious data placement** and evaluated cache-management methods against existing heuristics.

Technical Skills

Languages: Python, C/C++, CUDA, Go, Bash
ML and deployment: PyTorch, TensorFlow, ONNX, TorchScript
Systems: Kubernetes, Docker, Redis, PostgreSQL, Git, Prometheus
Research: ML theory, randomized algorithms, quantization, sketching, efficient attention

Service & Teaching

Reviewer: ICLR (2025 - 2026), ICML (2024 - 2026), NeurIPS (2024 - 2025).
External reviewer: FOCS (2025), CCCG (2023).
Teaching: NYU Basic Algorithms (Spring 2023); Algorithmic Machine Learning (Fall 2022).